

Education

Northeastern University, Boston, MA | Khoury College of Computer Sciences

Sep. 2022 - May 2026

B.S. in Computer Science and Game Development

GPA: 3.7 / 4.00 | Distinguished Scholar

Relevant Coursework: Object-Oriented Design | Algorithms and Data Structures | Game Programming 2 | Programming in C++ | Computer Systems | Computer Graphics | Network Fundamentals | Fundamentals in SWE | Game A.I.

Technical Skills

Programming Languages: TypeScript | JavaScript | Java | Python | C++ | C | SQL | Kotlin | GDScript | Assembly

Frameworks & Tools React | Node.js | Express | Vite | Next.js | Socket.io | Supabase | Firebase | Git/GitLab CI/CD | Vitest

Work/Experience

Research Engineer Intern | Sphere IoT

Jan. 2025 - Jun. 2025

- Built a remote IoT virtualization lab enabling off-site interaction with physical devices; provisioned and configured 100 devices.
- Designed and 3D printed modular mounts for button pushers using Fusion 360, ensuring compatibility across devices and documentation for future researchers.
- Implemented secure mTLS-based client-server communication for remote device control, enhancing security and reliability.
- Operated within a production GitLab CI/CD workflow, writing automated tests to validate changes and reduce regressions across networked services.

Developer | Scout

Jan. 2025 - Mar. 2025

- Developed the website for the club's annual conference (300+ attendees), ensuring accessibility compliance for users with disabilities.
 - Implemented accessibility features such as color adjustments for color blindness, text scalability, and keyboard-only navigation.
 - Collaborated with a cross-functional team to deliver an inclusive user experience for a diverse audience.
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Projects

Trackr | React, TypeScript, Express, [Socket.io](#), Vitest

Mar. 2026 - May 2026

- Built a modular local-first tracking app where users define and track anything (habits, moods, reps) from a single flexible schema; works fully offline via IndexedDB, then syncs to the cloud.
- Implemented cloud sync with Supabase and row-level-security auth, plus calendar and grid views, customizable per-tracker theming, and reusable templates.

Rebound Hound | 2D Artist & Gameplay Programmer

Jan. 2026 - May 2026

- Served as the sole 2D artist on a 5-person team, creating all pixel art sprites, animations, tilesets, shaders, and UI assets for a top-down action roguelike published on itch.io.
- Implemented character controllers and core mechanics including parrying and slingshotting; built procedural level generation that selects 9 of 13 rooms plus a final boss room.

GameNite | React, TypeScript, Express, [Socket.io](#), Vitest

Mar. 2026 - May. 2026

- Collaborated in a 4-person agile team to ship 3 features to an existing web platform; led frontend across all three.
 - Built a multiplayer Mafia social deduction game, implementing phase-based UI rendering, role-specific action controls, spectator mode, and win/loss logics using React and WebSockets.
 - Wrote automated tests across server and client using Vitest and React Testing Library.
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Additional Information

Certificate: Network Fundamentals

Spoken Languages: English (fluent), Korean (fluent), Japanese (conversational)